

The First Workshop on XR Accessibility

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1 INTRODUCTION

This workshop is the outgrowth of a Dagstuhl seminar (24371) held in September 2024 on *Extended Reality Accessibility*. The workshop brought together researchers, advocates, and other stakeholders to address concerns around accessibility in extended reality (XR). XR technologies are evolving rapidly, and have the potential to redefine communication and learning in the future. Ensuring that this evolution considers accessibility should be a primary design criterion for XR, as retroactive design is never as effective.

The Dagstuhl seminar attempted to bring together three communities that do not communicate well enough together to establish the importance of design criteria for future XR. The IEEE VR community contains researchers who lead the way in pioneering innovations in XR technology. XRAccess (xraccess.org) is a dedicated community advocating for XR environments that are not only state-of-the-art but universally accessible. The ASSETS and ACM SIGACCESS community are interested in advancing accessibility and design through innovative technologies that empower people with disabilities but are not focused on XR.

There are several dimensions of accessibility concerns for XR technologies. These include the following types: (1) Visual accessibility, e.g., color blindness, low vision/blindness, and text legibility; (2) auditory accessibility, e.g., deafness/hard of hearing, and spatial audio challenges; (3) physical/motor accessibility, e.g., limited mobility/dexterity, fatigue, and adaptive hardware limitations; and (4) cognitive accessibility, e.g., usability for cognitively or spatially challenged. Coupled with these are inconsistent or nonexistent accessibility features across XR technologies and limited developer awareness of the issues. How to address many of these concerns are open research problems, but will involve an interdisciplinary approach with participation of stakeholders in the relevant communities.

This workshop has a goal of engaging IEEE Virtual Reality attendees who are interested in but not yet working on XR Accessibility, and unsure where to start in terms of research questions and directions. As part of developing this workshop, we formed an XR Accessibility working group, which includes members from the recent Dagstuhl seminar and other stakeholders who participated in the workshop. The organizers are interested in broadening participation in this working group by bringing together people who are doing work in this area, broadly. A secondary goal of the workshop was for researchers who are already working on questions around

accessibility to have the opportunity to share their practices and results with others who may be interested in order to foster collaborations and future research projects.

2 PROGRAM

The XR Accessibility Workshop featured an engaging and interactive program aimed at inspiring and empowering participants to advance accessibility in extended realities. The agenda included an opening session where the organizers presented various aspects of XR accessibility that inspired the proposal for and development of the workshop. Time for discussion was given going into the poster presentations that showcased innovative ideas, solutions, and research related to XR accessibility in a collaborative and visual format. After the poster presentations, an interactive panel brought together key stakeholders from academia, industry, and advocacy to explore challenges, opportunities, and collaborative strategies for fostering inclusivity in XR. The stakeholders were interviewed by one of the workshop organizers (Dylan Fox) and asked to share their lived experiences, providing firsthand insights into the challenges faced by users with disabilities and emphasizing the need for accessible design in XR. Overall, the workshop emphasized dialogue and encouraged forging collaborations, offering a unique opportunity for participants to connect, learn, and contribute to shaping a more inclusive XR future.

3 REVIEWING PROCESS AND STATISTICS

The workshop called for submissions in the form of extended abstracts, resulting in ten received submissions. Each submission was reviewed by at least two members of our international program committee, which comprised 20 distinguished experts in the field of XR (see Section 5 for the full list). Following the initial review phase, authors were provided with feedback and asked to address the reviewers' comments and refine their submissions. Ultimately, we were able to accept nine submissions for poster presentations at the workshop.

4 OVERVIEW OF ACCEPTED ABSTRACTS

The workshop showcased nine extended abstracts that reflect the diversity of current research in accessible XR technologies. These contributions span topics such as biomechanical modeling, sensory augmentation, assistive technology, and inclusive design, providing valuable insights into the evolving field.

4.1 Biomechanical Modeling and Simulation

Two contributions explored biomechanical modeling to improve XR accessibility. The first [8] investigates how anthropometrically diverse biomechanical simulations can enhance design evaluation for mid-air interactions, revealing significant limitations in existing reinforcement learning models when applied to diverse user groups. The second [7] focuses on simulating central vision loss to study its effects on sensory cue usage during navigation tasks in virtual environments, offering insights into the role of multisensory integration for users with low vision.

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4.2 Sensory Augmentation and Assistive Technologies

Two works addressed the development of assistive technologies leveraging XR and AR. A novel system for VR shooting games [2] introduced an audio-based aiming mechanism to make gaming accessible to blind and visually impaired individuals, comparing various sonification methods. Meanwhile, the use of real-time 3D object detection in AR [1] was highlighted as a method to enhance object visibility and accessibility for users in diverse environments.

4.3 Accessibility for Individuals with Low Vision

Accessibility for individuals with low vision was a central theme in multiple studies. One study [5] examined visual attention in virtual environments for users with simulated and actual central vision loss, providing comparative insights using saliency maps and computational models. Another [6] investigated the impact of different obstacle-enhancement strategies in AR on walking performance, finding significant effects of augmentation type on mobility and user experience.

4.4 Inclusive and Accessible Design for Neurodiverse and Diverse User Populations

Inclusive design featured prominently in studies targeting neurodiverse populations and individuals with physical or cultural differences. A project [9] focused on using VR social stories to support neurodiverse students in educational settings, emphasizing the role of tailored, immersive environments in fostering skill-building and integration. A similar emphasis on customization was evident in a study [3] that explored methods for adapting virtual museum experiences to individual user needs, ensuring accessibility across sensory, motor, and cultural dimensions. Additionally, the potential of XR systems to support neurodegenerative disease patients was highlighted in a case study [4] on Parkinson's disease, showcasing multimodal guidance and task calibration to enhance usability and engagement.

5 INTERNATIONAL PROGRAM COMMITTEE

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- Lauren Buck, University of Utah, United States
- Sarah Creem-Regehr, University of Utah, United States
- Tiare Feuchtner, University of Konstanz, Germany
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